🔍Linux `grep` Command Cheat Sheet

`grep` = **Global Regular Expression Print**

👉Used to **search text patterns** inside files or command outputs.

📌Basic Usage

```
grep "word" file.txt   # Search for a word in a file
```

Example 1

File (`file.txt`):

```
This is a sample text file.
It contains multiple lines.
We are searching for a specific word.
```

```
$ grep "word" file.txt
We are searching for a specific word.
```

Example 2

File (`file.txt`):

```
Hello World
This is Linux
Welcome to grep
```

```
$ grep "Linux" file.txt
This is Linux
```

⚙️Common Options with Examples

1. Case-insensitive (`-i`)

```
$ grep -i "grep" file.txt
welcome to GREP
```

2. Show Line Numbers (`-n`)

```
$ grep -n "Error" file.txt
1:Error: File not found
3:Error: Access denied
```

3. Invert Match (`-v`)

```
$ grep -v "Error" file.txt
Success: Operation completed
Warning: Low disk space
```

4. Count Matches (`-c`)

```
$ grep -c "fail" file.txt
3
```

5. Show Filenames with Matches (`-l`)

```
$ grep -l "Universe" *.txt
file2.txt
```

6. Recursive Search (`-r`)

```
$ grep -r "main" /project/src
/project/src/app.java: public static void main(String[] args) {
/project/src/utils.java: public static void main(String[] args) {
```

7. Match Whole Words Only (`-w`)

```
$ grep -w "cat" file.txt
The cat is on the roof.
```

8. Print Only Matches (`-o`)

```
$ grep -o "foo[0-9]" file.txt
foo1
foo3
```

9. Show Context (`-A`, `-B`, `-C`)

```
$ grep -A 2 "Error" file.txt   # After
Error occurred here
Line 3
Line 4

$ grep -B 2 "Error" file.txt   # Before
Line 1
Line 2
Error occurred here

$ grep -C 2 "Error" file.txt   # Context (before+after)
Line 1
Line 2
Error occurred here
Line 4
Line 5
```

📖Regex Examples with `grep`

1. Match Lines Starting or Ending

```
$ grep ^"Start" file.txt   # Lines starting with "Start"
Start the process now
Start again

$ grep "End$" file.txt     # Lines ending with "End"
Ending soon
```

2. Match Pattern (e.g., SSN format)

```
$ grep "[0-9]\{3\}-[0-9]\{2\}-[0-9]\{4\}" file.txt
My SSN is 123-45-6789
Another one: 987-65-4321
```

3. Extended Regex (`-E`)

```
$ grep -E "cat|dog" file.txt
I have a cat.
You have a dog.
```

4. Fixed String (`-F`)

```
$ grep -F "exact string" file.txt
This line contains exact string
```

👉`-F` treats text as a **literal string**, not regex.

📊Summary Table

Option	Meaning	Example
`-i`	Case-insensitive	`grep -i "hello" file.txt`
`-n`	Show line numbers	`grep -n "error" logfile.log`
`-v`	Invert match	`grep -v "success" file.txt`
`-c`	Count matches	`grep -c "fail" file.txt`
`-l`	Show filenames with matches	`grep -l "TODO" *.java`
`-r`	Recursive search	`grep -r "main" ./src`
`-w`	Match whole words	`grep -w "cat" file.txt`
`-o`	Show only matches	`grep -o "foo[0-9]" file.txt`
`-A n`	Show n lines after match	`grep -A 2 "Error" file.txt`
`-B n`	Show n lines before match	`grep -B 2 "Error" file.txt`
`-C n`	Show n lines around match	`grep -C 2 "Error" file.txt`

👉This cheat sheet covers **basic searching, options, regex, and real examples** for `grep`.